Conduction current, which is discussed in more detail in Section 4-6, obeys Ohm's law, whereas convection current does not.

Concept Question 4-1: What happens to Maxwell's equations under static conditions?

Concept Question 4-2: How is the current density **J** related to the volume charge density ρ_v?

Concept Question 4-3: What is the difference between convection and conduction currents?

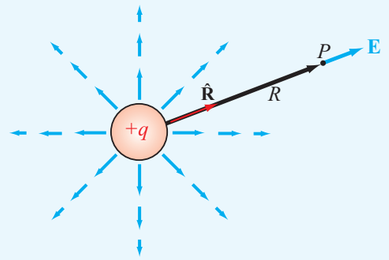


Figure 4-3 Electric field lines due to a charge q.

4-3 Coulomb's Law

One of the primary goals of this chapter is to develop dexterity in applying the expressions for the ***electric field intensity*** **E** and associated ***electric flux density*** **D** induced by a specified distribution of charge. Our discussion will be limited to electrostatic fields induced by stationary charge densities.

We begin by reviewing the expression for the electric field introduced in Section 1-3.2 on the basis of the results of Coulomb's experiments on the electrical force between charged bodies. ***Coulomb's law***, which was first introduced for electrical charges in air and later generalized to material media, implies that:

(1) An isolated charge q induces an electric field **E** at every point in space, and at any specific point P, **E** is given by

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\varepsilon R^2} \quad \text{(V/m)}, \tag{4.13}$$

where $\hat{\mathbf{R}}$ is a unit vector pointing from q to P (**Fig. 4-3**), R is the distance between them, and ε is the electrical permittivity of the medium containing the observation point P.

(2) In the presence of an electric field **E** at a given point in space that may be due to a single charge or a distribution of charges, the force acting on a test charge q' when placed at P is

$$\mathbf{F} = q'\mathbf{E} \quad \text{(N)}. \tag{4.14}$$

With **F** measured in newtons (N) and q' in coulombs (C), the unit of **E** is (N/C), which will be shown later in Section 4-5 to be the same as volt per meter (V/m).

For a material with electrical permittivity ε, the electric field quantities **D** and **E** are related by

$$\mathbf{D} = \varepsilon\mathbf{E} \tag{4.15}$$

with

$$\varepsilon = \varepsilon_r\varepsilon_0, \tag{4.16}$$

where

$$\varepsilon_0 = 8.85 \times 10^{-12} \approx (1/36\pi) \times 10^{-9} \quad \text{(F/m)}$$

is the electrical permittivity of free space and $\varepsilon_r = \varepsilon/\varepsilon_0$ is called the ***relative permittivity*** (or ***dielectric constant***) of the material. For most materials and under a wide range of conditions, ε is independent of both the magnitude and direction of **E** [as implied by Eq. (4.15)].

▶ If ε is independent of the magnitude of **E**, then the material is said to be ***linear*** because **D** and **E** are related linearly, and if it is independent of the direction of **E**, the material is said to be ***isotropic.*** ◀

Materials usually do not exhibit nonlinear permittivity behavior except when the amplitude of **E** is very high (at levels approaching dielectric breakdown conditions discussed later in Section 4-7), and anisotropy is present only in certain materials with peculiar crystalline structures. Hence, except for unique materials under very special circumstances, the quantities **D** and **E** are effectively redundant; for a material with known ε, knowledge of either **D** or **E** is sufficient to specify the other in that material.

4-3.1 Electric Field Due to Multiple Point Charges

The expression given by Eq. (4.13) for the field **E** due to a single point charge can be extended to multiple charges. We begin by considering two point charges, q_1 and q_2, with position vectors $\mathbf{R}_1$ and $\mathbf{R}_2$ (measured from the origin in

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传导电流（将在第 4-6 节中详细讨论）遵守欧姆定律，而对流电流则不然。

Concept Question 4-1: What happens to Maxwell's equations under static conditions?

Concept Question 4-2: How is the current density **J** related to the volume charge density ρ_v?

Concept Question 4-3: What is the difference between convection and conduction currents?

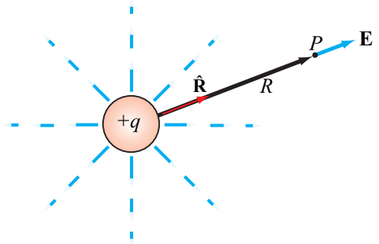


图 4-3 电荷 q 产生的电场线。

4-3 库仑定律

本章的主要目标之一是提高应用特定电荷分布引起的电场强度 E 和相关电通量密度 D 表达式的灵活性。我们的讨论将仅限于由静止电荷密度引起的静电场。

我们首先根据库仑关于带电体之间电力的实验结果回顾 1-3.2 节中介绍的电场表达式。库仑定律首先针对空气中的电荷引入，后来推广到材料介质，它意味着：

(1) 孤立电荷 q 在空间中的每一点感应出一个电场 E，并且在任何特定点 P，E 由下式给出

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\varepsilon R^2} \quad \text{(V/m)}, \tag{4.13}$$

其中R是从q指向P的单位向量（图4-3），R是它们之间的距离，ε是包含观察点P的介质的介电常数。

(2) 当空间中给定点存在电场 E（可能是由于单个电荷或电荷分布）时，当放置在 P 处时，作用在测试电荷 q′ 上的力为

$$\mathbf{F} = q'\mathbf{E} \quad \text{(N)}. \tag{4.14}$$

F 的单位为牛顿 (N)，q′ 的单位为库仑 (C)，E 的单位为 (N/C)，稍后将在第 4-5 节中显示，它与伏每米 (V/m) 相同。

对于介电常数为 ε 的材料，电场量 D 和 E 的关系为

$$\mathbf{D} = \varepsilon\mathbf{E} \tag{4.15}$$

和

$$\varepsilon = \varepsilon_r\varepsilon_0, \tag{4.16}$$

在哪里

$$\varepsilon_0 = 8.85 \times 10^{-12} \approx (1/36\pi) \times 10^{-9} \quad \text{(F/m)}$$

是自由空间的介电常数，$\varepsilon_r = \varepsilon/\varepsilon_0$ 称为材料的相对介电常数（或介电常数）。对于大多数材料和在各种条件下，ε 与 E 的大小和方向无关[如方程 1 所暗示的那样]。 (4.15)]。

► 如果 ε 与 E 的大小无关，则该材料被称为线性，因为 D 和 E 线性相关；如果它与 E 的方向无关，则该材料被称为各向同性。 ◄

材料通常不会表现出非线性介电常数行为，除非 E 的幅度非常高（接近后面第 4-7 节讨论的介电击穿条件的水平），并且各向异性仅存在于某些具有特殊晶体结构的材料中。因此，除了非常特殊情况下的独特材料外，数量D和E实际上是多余的；对于已知 ε 的材料，D 或 E 的知识足以指定该材料中的另一个。

4-3.1 多点电荷产生的电场

方程给出的表达式。 (4.13) 对于场E，由于单点电荷可以扩展到多个电荷。我们首先考虑两个点电荷 q_1 和 q_2，位置向量 R_1 和 R_2 （从原点测量

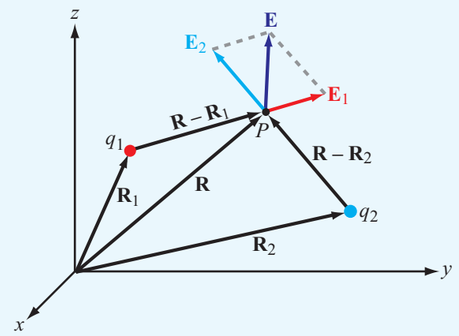


Figure 4-4 The electric field **E** at *P* due to two charges is equal to the vector sum of $\mathbf{E}_1$ and $\mathbf{E}_2$.

Fig. 4-4). The electric field **E** is to be evaluated at a point *P* with position vector **R**. At *P*, the electric field $\mathbf{E}_1$ due to q_1 alone is given by Eq. (4.13) with *R*, which is the distance between q_1 and *P*, replaced with $|\mathbf{R}-\mathbf{R}_1|$ and the unit vector $\hat{\mathbf{R}}$ replaced with $(\mathbf{R}-\mathbf{R}_1)/|\mathbf{R}-\mathbf{R}_1|$. Thus,

$$\mathbf{E}_1 = \frac{q_1(\mathbf{R}-\mathbf{R}_1)}{4\pi\varepsilon|\mathbf{R}-\mathbf{R}_1|^3} \quad \text{(V/m)}. \qquad (4.17a)$$

Similarly, the electric field at *P* due to q_2 alone is

$$\mathbf{E}_2 = \frac{q_2(\mathbf{R}-\mathbf{R}_2)}{4\pi\varepsilon|\mathbf{R}-\mathbf{R}_2|^3} \quad \text{(V/m)}. \qquad (4.17b)$$

► The electric field obeys the principle of linear superposition. ◄

Hence, the total electric field **E** at *P* due to q_1 and q_2 together is determined as

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = \frac{1}{4\pi\varepsilon}\left[\frac{q_1(\mathbf{R}-\mathbf{R}_1)}{|\mathbf{R}-\mathbf{R}_1|^3} + \frac{q_2(\mathbf{R}-\mathbf{R}_2)}{|\mathbf{R}-\mathbf{R}_2|^3}\right]. \qquad (4.18)$$

Generalizing the preceding result to the case of *N* point charges, the electric field **E** at point *P* with position vector **R** due to charges $q_1, q_2, \ldots, q_N$ located at points with position vectors $\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_N$ equals the vector sum of the electric fields induced by all the individual charges. Thus,

$$\mathbf{E} = \frac{1}{4\pi\varepsilon}\sum_{i=1}^{N}\frac{q_i(\mathbf{R}-\mathbf{R}_i)}{|\mathbf{R}-\mathbf{R}_i|^3} \quad \text{(V/m)}. \qquad (4.19)$$

Example 4-3: Electric Field Due to Two Point Charges

Two point charges with

$$q_1 = 2\times10^{-5}\ \text{C}$$

and

$$q_2 = -4\times10^{-5}\ \text{C}$$

are located in free space at points with Cartesian coordinates $(1,3,-1)$ and $(-3,1,-2)$, respectively. Find (a) the electric field **E** at $(3,1,-2)$ and (b) the force on a 8×10^{-5} C charge located at that point. All distances are in meters.

Solution: (a) From Eq. (4.18), the electric field **E** with $\varepsilon = \varepsilon_0$ (free space) is

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\left[q_1\frac{(\mathbf{R}-\mathbf{R}_1)}{|\mathbf{R}-\mathbf{R}_1|^3} + q_2\frac{(\mathbf{R}-\mathbf{R}_2)}{|\mathbf{R}-\mathbf{R}_2|^3}\right] \quad \text{(V/m)}.$$

The vectors $\mathbf{R}_1$, $\mathbf{R}_2$, and **R** are

$$\mathbf{R}_1 = \hat{\mathbf{x}}+\hat{\mathbf{y}}3-\hat{\mathbf{z}},$$
$$\mathbf{R}_2 = -\hat{\mathbf{x}}3+\hat{\mathbf{y}}-\hat{\mathbf{z}}2,$$
$$\mathbf{R} = \hat{\mathbf{x}}3+\hat{\mathbf{y}}-\hat{\mathbf{z}}2.$$

Hence,

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\left[\frac{2(\hat{\mathbf{x}}2-\hat{\mathbf{y}}2-\hat{\mathbf{z}})}{27} - \frac{4(\hat{\mathbf{x}}6)}{216}\right]\times10^{-5} = \frac{\hat{\mathbf{x}}-\hat{\mathbf{y}}4-\hat{\mathbf{z}}2}{108\pi\varepsilon_0}\times10^{-5} \quad \text{(V/m)}.$$

(b) The force on q_3 is

$$\mathbf{F} = q_3\mathbf{E} = 8\times10^{-5}\times\frac{\hat{\mathbf{x}}-\hat{\mathbf{y}}4-\hat{\mathbf{z}}2}{108\pi\varepsilon_0}\times10^{-5} = \frac{\hat{\mathbf{x}}2-\hat{\mathbf{y}}8-\hat{\mathbf{z}}4}{27\pi\varepsilon_0}\times10^{-10} \quad \text{(N)}.$$

Exercise 4-3: Four charges of 10 μC each are located in free space at points with Cartesian coordinates $(-3,0,0)$, $(3,0,0)$, $(0,-3,0)$, and $(0,3,0)$. Find the force on a 20 μC charge located at $(0,0,4)$. All distances are in meters.

Answer: $\mathbf{F} = \hat{\mathbf{z}}0.23$ N. (See EM.)

Exercise 4-4: Two identical charges are located on the *x* axis at $x=3$ and $x=7$. At what point in space is the net electric field zero?

Answer: At point $(5,0,0)$. (See EM.)

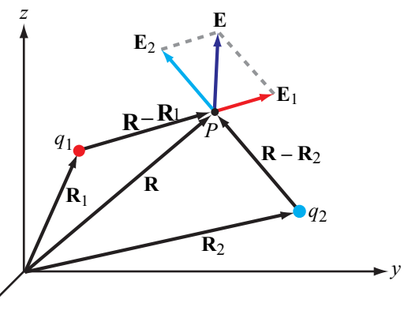


图 4-4 P 处两个电荷产生的电场 E 等于 E_1 和 E_2 的矢量和。

图4-4）。电场 E 将在具有位置矢量 R 的点 P 处进行评估。在 P 处，仅由 q_1 产生的电场 E_1 由方程 1 给出。(4.13) 将 R（即 q_1 和 P 之间的距离）替换为 $|R - R_1|$，单位向量 $\hat{R}$ 替换为 $(R - R_1)/|R - R_1|$。因此，

$$\mathbf{E}_1 = \frac{q_1(\mathbf{R}-\mathbf{R}_1)}{4\pi\varepsilon|\mathbf{R}-\mathbf{R}_1|^3} \quad \text{(V/m)}. \qquad (4.17a)$$

类似地，仅由 q_2 产生的 P 处电场为

$$\mathbf{E}_2 = \frac{q_2(\mathbf{R}-\mathbf{R}_2)}{4\pi\varepsilon|\mathbf{R}-\mathbf{R}_2|^3} \quad \text{(V/m)}. \qquad (4.17b)$$

► 电场遵循线性叠加原理
位置。◄

因此，由 q_1 和 q_2 共同产生的 P 处的总电场 E 确定为

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = \frac{1}{4\pi\varepsilon}\left[\frac{q_1(\mathbf{R}-\mathbf{R}_1)}{|\mathbf{R}-\mathbf{R}_1|^3} + \frac{q_2(\mathbf{R}-\mathbf{R}_2)}{|\mathbf{R}-\mathbf{R}_2|^3}\right]. \qquad (4.18)$$

将前面的结果推广到 N 个点电荷的情况，由于位于位置矢量 R_1、R_2、…、R_N 的点上的电荷 q_1、q_2、…、q_N，在位置矢量 R 的 P 点处的电场 E 等于所有单个电荷感应的电场的矢量和。因此，

$$\mathbf{E} = \frac{1}{4\pi\varepsilon}\sum_{i=1}^{N}\frac{q_i(\mathbf{R}-\mathbf{R}_i)}{|\mathbf{R}-\mathbf{R}_i|^3} \quad \text{(V/m)}。 \qquad (4.19)$$

例 4-3： 两点电荷产生的电场

两点电荷

$$q_1 = 2\times10^{-5}\ \text{C}$$

和

$$q_2 = -4\times10^{-5}\ \text{C}$$

分别位于自由空间中具有笛卡尔坐标 (1, 3, −1) 和 (−3, 1, −2) 的点处。求 (a) (3, 1, −2) 处的电场 E 和 (b) 位于该点的 8×10^{-5} C 电荷上的力。所有距离均以米为单位。

解： (a) 根据方程。 （4.18），$\varepsilon = \varepsilon_0$ (自由空间)的电场E为

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\left[q_1\frac{(\mathbf{R}-\mathbf{R}_1)}{|\mathbf{R}-\mathbf{R}_1|^3} + q_2\frac{(\mathbf{R}-\mathbf{R}_2)}{|\mathbf{R}-\mathbf{R}_2|^3}\right] \quad \text{(V/m)}.$$

向量 R_1、R_2 和 R 是

$$\mathbf{R}_1 = \hat{\mathbf{x}}+\hat{\mathbf{y}}3-\hat{\mathbf{z}},$$
$$\mathbf{R}_2 = -\hat{\mathbf{x}}3+\hat{\mathbf{y}}-\hat{\mathbf{z}}2,$$
$$\mathbf{R} = \hat{\mathbf{x}}3+\hat{\mathbf{y}}-\hat{\mathbf{z}}2.$$

因此，

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\left[\frac{2(\hat{\mathbf{x}}2-\hat{\mathbf{y}}2-\hat{\mathbf{z}})}{27} - \frac{4(\hat{\mathbf{x}}6)}{216}\right]\times10^{-5} = \frac{\hat{\mathbf{x}}-\hat{\mathbf{y}}4-\hat{\mathbf{z}}2}{108\pi\varepsilon_0}\times10^{-5} \quad \text{(V/m)}.$$

(b) q_3 上的力为

$$\mathbf{F} = q_3\mathbf{E} = 8\times10^{-5}\times\frac{\hat{\mathbf{x}}-\hat{\mathbf{y}}4-\hat{\mathbf{z}}2}{108\pi\varepsilon_0}\times10^{-5} = \frac{\hat{\mathbf{x}}2-\hat{\mathbf{y}}8-\hat{\mathbf{z}}4}{27\pi\varepsilon_0}\times10^{-10} \quad \text{(N)}.$$

练习 4-3：四个 10 μ C 的电荷位于自由空间中笛卡尔坐标为 (−3, 0, 0)、(3, 0, 0)、(0, −3, 0) 和 (0, 3, 0) 的点处。求 (0, 0, 4) 处 20 μ C 电荷所受的力。所有距离均以米为单位。

Answer: $\mathbf{F} = \hat{\mathbf{z}}0.23$ N. (See EM.)

练习4-4：两个相同的电荷位于x 轴上的$x = 3$ 和$x = 7$ 处。空间中的什么点净电场为零？

答案：在点(5,0,0)。 (参见EM。)

Exercise 4-5: In a hydrogen atom, the electron and proton are separated by an average distance of 5.3×10^{-11} m. Find the magnitude of the electrical force F_e between the two particles, and compare it with the gravitational force F_g between them.

Answer: $F_e = 8.2 \times 10^{-8}$ N, and $F_g = 3.6 \times 10^{-47}$ N. (See EM.)

4-3.2 Electric Field Due to a Charge Distribution

We now extend the results obtained for the field due to discrete point charges to continuous charge distributions. Consider a volume υ' that contains a distribution of electric charge with volume charge density ρ_v, which may vary spatially within υ' (Fig. 4-5). The differential electric field at a point P due to a differential amount of charge $dq = \rho_v\, d\upsilon'$ contained in a differential volume $d\upsilon'$ is

$$d\mathbf{E} = \hat{\mathbf{R}}' \frac{dq}{4\pi\varepsilon R'^2} = \hat{\mathbf{R}}' \frac{\rho_v\, d\upsilon'}{4\pi\varepsilon R'^2}, \tag{4.20}$$

where $\mathbf{R}'$ is the vector from the differential volume $d\upsilon'$ to point P. Applying the principle of linear superposition, the total electric field $\mathbf{E}$ is obtained by integrating the fields due to all differential charges in υ'. Thus,

$$\mathbf{E} = \int_{\upsilon'} d\mathbf{E} = \frac{1}{4\pi\varepsilon} \int_{\upsilon'} \hat{\mathbf{R}}' \frac{\rho_v\, d\upsilon'}{R'^2}. \tag{4.21a}$$

(volume distribution)

It is important to note that, in general, both R' and $\hat{\mathbf{R}}'$ vary as a function of position over the integration volume υ'.

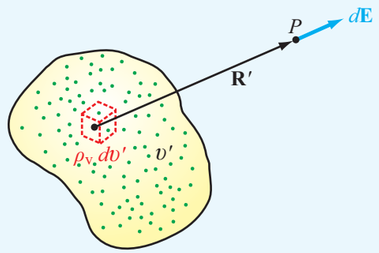


Figure 4-5 Electric field due to a volume charge distribution.

If the charge is distributed across a surface S' with surface charge density ρ_s, then $dq = \rho_s\, ds'$, and if it is distributed along a line l' with a line charge density ρ_ℓ, then $dq = \rho_\ell\, dl'$. Accordingly, the electric fields due to surface and line charge distributions are

$$\mathbf{E} = \frac{1}{4\pi\varepsilon} \int_{S'} \hat{\mathbf{R}}' \frac{\rho_s\, ds'}{R'^2}, \tag{4.21b}$$

(surface distribution)

$$\mathbf{E} = \frac{1}{4\pi\varepsilon} \int_{l'} \hat{\mathbf{R}}' \frac{\rho_\ell\, dl'}{R'^2}. \tag{4.21c}$$

(line distribution)

Example 4-4: Electric Field of a Ring of Charge

A ring of charge of radius b is characterized by a uniform line charge density of positive polarity ρ_ℓ. The ring resides in free space and is positioned in the x–y plane, as shown in Fig. 4-6. Determine the electric field intensity $\mathbf{E}$ at a point $P = (0,0,h)$ along the axis of the ring at a distance h from its center.

Solution: We start by considering the electric field generated by a differential ring segment with cylindrical coordinates $(b,\phi,0)$ in Fig. 4-6(a). The segment has length $dl = b\, d\phi$ and contains charge $dq = \rho_\ell\, dl = \rho_\ell b\, d\phi$. The distance vector $\mathbf{R}'_1$ from segment 1 to point $P = (0,0,h)$ is

$$\mathbf{R}'_1 = -\hat{\mathbf{r}}b + \hat{\mathbf{z}}h,$$

from which it follows that

$$R'_1 = |\mathbf{R}'_1| = \sqrt{b^2+h^2}, \qquad \hat{\mathbf{R}}'_1 = \frac{\mathbf{R}'_1}{|\mathbf{R}'_1|} = \frac{-\hat{\mathbf{r}}b + \hat{\mathbf{z}}h}{\sqrt{b^2+h^2}}.$$

The electric field at $P = (0,0,h)$ due to the charge in segment 1 therefore is

$$d\mathbf{E}_1 = \frac{1}{4\pi\varepsilon_0} \hat{\mathbf{R}}'_1 \frac{\rho_\ell\, dl}{R'^2_1} = \frac{\rho_\ell b}{4\pi\varepsilon_0} \frac{(-\hat{\mathbf{r}}b + \hat{\mathbf{z}}h)}{(b^2+h^2)^{3/2}}\, d\phi.$$

The field $d\mathbf{E}_1$ has component dE_{1r} along $-\hat{\mathbf{r}}$ and component dE_{1z} along $\hat{\mathbf{z}}$. From symmetry considerations, the field $d\mathbf{E}_2$ generated by differential segment 2 in Fig. 4-6(b), which is located diametrically opposite to segment 1, is identical to $d\mathbf{E}_1$ except that the $\hat{\mathbf{r}}$ component of $d\mathbf{E}_2$ is opposite that

练习 4-5：在氢原子中，电子和质子的平均距离为 5.3 × 10^{-11} m。求两个粒子之间的电力 F_e 的大小，并将其与它们之间的重力 F_g 进行比较。

答案：$F_e = 8.2 \times 10^{-8}$ N，和 $F_g = 3.6 \times 10^{-47}$ N。（参见 EM 。）

4-3.2 电荷分布产生的电场

现在，我们将由于离散点电荷而获得的结果扩展到连续电荷分布。考虑一个体积 υ′，它包含具有体积电荷密度 ρ_v 的电荷分布，其可能在 υ′（内随空间变化。 4-5)。由于微分体积 dυ′ 中包含的微分电荷量 $dq = \rho_v\, d\upsilon'$，在点 P 处的微分电场为

$$d\mathbf{E} = \hat{\mathbf{R}}' \frac{dq}{4\pi\varepsilon R'^2} = \hat{\mathbf{R}}' \frac{\rho_v\, d\upsilon'}{4\pi\varepsilon R'^2}, \tag{4.20}$$

其中R′是微分体积dυ′到点P的矢量。应用线性叠加原理，对υ′中所有微分电荷产生的电场进行积分，得到总电场E。因此，

$$\mathbf{E} = \int_{\upsilon'} d\mathbf{E} = \frac{1}{4\pi\varepsilon} \int_{\upsilon'} \hat{\mathbf{R}}' \frac{\rho_v\, d\upsilon'}{R'^2}. \tag{4.21a}$$

(volume distribution)

值得注意的是，一般来说，R′ 和 $\hat{\mathrm{R}}'$ 都随着积分体积 υ′ 上的位置而变化。

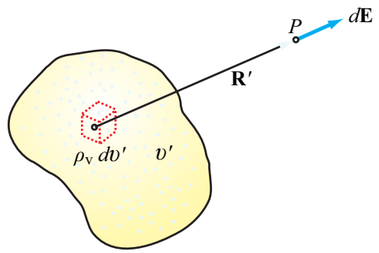


图 4-5 体积电荷分布产生的电场。

如果电荷以表面电荷密度 ρ_s 分布在表面 S′ 上，则为 dq = ρ_s ds′；如果电荷以线电荷密度 ρ_ℓ 沿线 l′ 分布，则为 dq = ρ_ℓ dl′。因此，由于表面和线电荷分布而产生的电场为

$$\mathbf{E} = \frac{1}{4\pi\varepsilon} \int_{S'} \hat{\mathbf{R}}' \frac{\rho_s\, ds'}{R'^2}, \tag{4.21b}$$

(表面分布)

$$\mathbf{E} = \frac{1}{4\pi\varepsilon} \int_{l'} \hat{\mathbf{R}}' \frac{\rho_\ell\, dl'}{R'^2}. \tag{4.21c}$$

(line distribution)

例 4-4：电荷环的电场

半径为 b 的电荷环的特征是均匀的正极性线电荷密度 ρ_ℓ。该环位于自由空间中，位于 x-y 平面中，如图 4-6 所示。确定沿环轴距环中心距离 h 的点 P = (0, 0, h) 处的电场强度 E。

解：我们首先考虑图4-6（a）中柱坐标（b，ϕ，0）的微分环段产生的电场。该段的长度为 dl = b dϕ，并包含电荷 dq = ρ_ℓ dl = ρ_ℓb dϕ。从线段 1 到点 P = (0, 0, h) 的距离向量 R'_1 为

$$\mathbf{R}'_1 = -\hat{\mathbf{r}}b + \hat{\mathbf{z}}h,$$

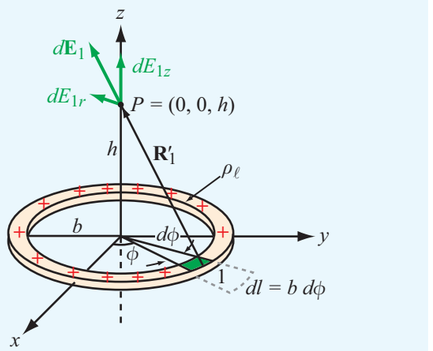
由此可见

$$R'_1 = |\mathbf{R}'_1| = \sqrt{b^2+h^2}, \qquad \hat{\mathbf{R}}'_1 = \frac{\mathbf{R}'_1}{|\mathbf{R}'_1|} = \frac{-\hat{\mathbf{r}}b + \hat{\mathbf{z}}h}{\sqrt{b^2+h^2}}.$$

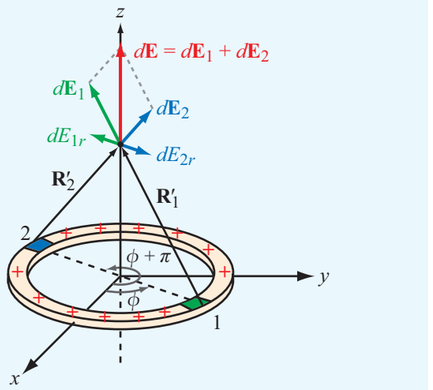
因此，由于段 1 中的电荷，P = (0, 0, h) 处的电场为

$$d\mathbf{E}_1 = \frac{1}{4\pi\varepsilon_0} \hat{\mathbf{R}}'_1 \frac{\rho_\ell\, dl}{R'^2_1} = \frac{\rho_\ell b}{4\pi\varepsilon_0} \frac{(-\hat{\mathbf{r}}b + \hat{\mathbf{z}}h)}{(b^2+h^2)^{3/2}}\, d\phi.$$

域 dE_1 具有沿 $-\hat{\mathrm{r}}$ 的分量 dE_{1r} 和沿 $\hat{\mathrm{z}}$ 的分量 dE_{1z}。从对称性考虑，图 4-6(b) 中与线段 1 完全相反的微分线段 2 生成的场 dE_2 与 dE_1 相同，只是 dE_2 的 $\hat{\mathrm{r}}$ 分量与



(a)



(b)

Figure 4-6 Ring of charge with line density ρ_ℓ. (a) The field $d\mathbf{E}_1$ due to infinitesimal segment 1 and (b) the fields $d\mathbf{E}_1$ and $d\mathbf{E}_2$ due to segments at diametrically opposite locations (Example 4-4).

opposite at $(\phi + \pi)$, we can obtain the total field generated by the ring by integrating Eq. (4.22) over a semicircle as

$$\mathbf{E} = \hat{\mathbf{z}}\,\frac{\rho_\ell bh}{2\pi\varepsilon_0(b^2+h^2)^{3/2}}\int_0^{\pi} d\phi$$

$$= \hat{\mathbf{z}}\,\frac{\rho_\ell bh}{2\varepsilon_0(b^2+h^2)^{3/2}} = \hat{\mathbf{z}}\,\frac{h}{4\pi\varepsilon_0(b^2+h^2)^{3/2}}\,Q, \qquad (4.23)$$

where $Q = 2\pi b\rho_\ell$ is the total charge on the ring.

Example 4-5: Electric Field of a Circular Disk of Charge

Find the electric field at point P with Cartesian coordinates $(0,0,h)$ due to a circular disk of radius a and uniform charge density ρ_s residing in the x–y plane (Fig. 4-7). Also, evaluate $\mathbf{E}$ due to an infinite sheet of charge density ρ_s by letting $a \to \infty$.

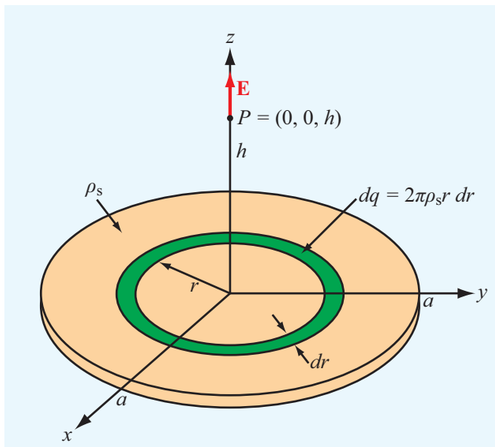


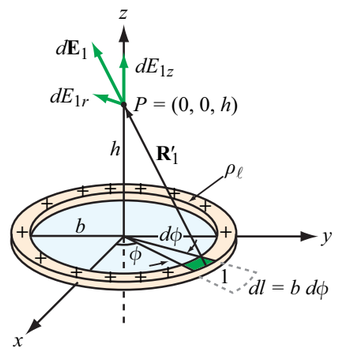
Figure 4-7 Circular disk of charge with surface charge density ρ_s. The electric field at $P = (0,0,h)$ points along the z direction (Example 4-5).

of $d\mathbf{E}_1$. Hence, the $\hat{\mathbf{r}}$ components in the sum cancel and the $\hat{\mathbf{z}}$ contributions add. The sum of the two contributions is

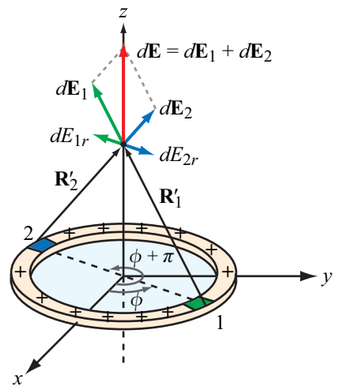
$$d\mathbf{E} = d\mathbf{E}_1 + d\mathbf{E}_2 = \hat{\mathbf{z}}\,\frac{\rho_\ell bh}{2\pi\varepsilon_0}\,\frac{d\phi}{(b^2+h^2)^{3/2}}. \qquad (4.22)$$

Since for every ring segment in the semicircle defined over the azimuthal range $0 \le \phi \le \pi$ (the right-hand half of the circular ring) there is a corresponding segment located diametrically

Solution: Building on the expression obtained in Example 4-4 for the on-axis electric field due to a circular ring of charge, we can determine the field due to the circular disk by treating the disk as a set of concentric rings. A ring of radius r and width dr has an area $ds = 2\pi r\,dr$ and contains charge $dq = \rho_s\,ds = 2\pi\rho_s r\,dr$. Upon using this expression in



(a)



(b)

图 4-6 具有线密度 ρ_ℓ 的电荷环。(a) 由于无穷小线段 1 产生的场 dE_1，(b) 由于位于完全相反位置的线段而产生的场 dE_1 和 dE_2（例 4-4）。

相反，在$(\phi+\pi)$处，我们可以通过积分方程得到环产生的总场。(4.22) 在半圆上为

$$\mathbf{E} = \hat{\mathbf{z}}\,\frac{\rho_\ell bh}{2\pi\varepsilon_0(b^2+h^2)^{3/2}}\int_0^{\pi} d\phi$$

$$= \hat{\mathbf{z}}\,\frac{\rho_\ell bh}{2\varepsilon_0(b^2+h^2)^{3/2}} = \hat{\mathbf{z}}\,\frac{h}{4\pi\varepsilon_0(b^2+h^2)^{3/2}}\,Q, \qquad (4.23)$$

其中 $Q = 2\pi b\rho_\ell$ 是环上的总电荷。

例 4-5：带电圆盘的电场

求由于半径为 a 的圆盘和 x-y 平面内均匀电荷密度 ρ_s 导致的笛卡尔坐标 (0, 0, h) 处 P 点的电场（图 4-7）。另外，通过设 $\to \infty$ 来评估由于无限电荷密度 ρ_s 引起的 E。

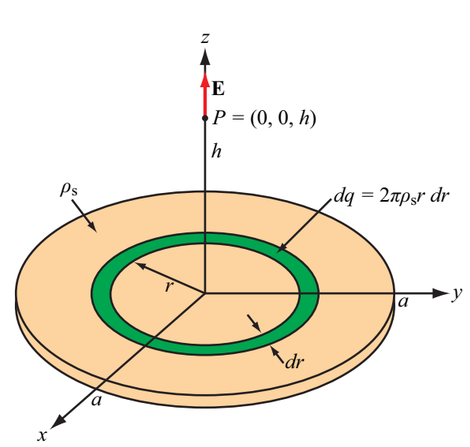


图 4-7 表面电荷密度为 ρ_s 的电荷圆盘。$P = (0,0,h)$ 处的电场沿 z 方向指向（示例 4-5）。

dE_1。因此，总和中的 $\hat{r}$ 分量抵消，而 $\hat{z}$ 贡献相加。两个贡献的总和是

$$d\mathbf{E} = d\mathbf{E}_1 + d\mathbf{E}_2 = \hat{\mathbf{z}}\,\frac{\rho_\ell bh}{2\pi\varepsilon_0}\,\frac{d\phi}{(b^2+h^2)^{3/2}}. \qquad (4.22)$$

因为对于在方位角范围 $0 \le \phi \le \pi$ (圆环的右半部分) 上定义的半圆中的每个环段，都有一个位于直径方向的相应段

解：根据例 4-4 中获得的关于圆环电荷产生的轴上电场的表达式，我们可以通过将圆盘视为一组同心环来确定圆盘产生的电场。半径为 r、宽度为 dr 的环的面积为 $ds = 2\pi r\,dr$，并且包含电荷 $dq = \rho_s\,ds = 2\pi\rho_s r\,dr$。在使用这个表达式时

Eq. (4.23) and also replacing b with r, we obtain the following expression for the field due to the ring:

$$d\mathbf{E} = \hat{\mathbf{z}}\,\frac{h}{4\pi\varepsilon_0(r^2+h^2)^{3/2}}\,(2\pi\rho_s r\,dr).$$

The total field at P is obtained by integrating the expression over the limits $r = 0$ to $r = a$:

$$\mathbf{E} = \hat{\mathbf{z}}\,\frac{\rho_s h}{2\varepsilon_0}\int_0^a \frac{r\,dr}{(r^2+h^2)^{3/2}} = \pm\hat{\mathbf{z}}\,\frac{\rho_s}{2\varepsilon_0}\left[1-\frac{|h|}{\sqrt{a^2+h^2}}\right], \tag{4.24}$$

with the plus sign for $h > 0$ (P above the disk) and the minus sign when $h < 0$ (P below the disk).

For an infinite sheet of charge with $a = \infty$,

$$\mathbf{E} = \pm\hat{\mathbf{z}}\,\frac{\rho_s}{2\varepsilon_0}. \tag{4.25}$$

(infinite sheet of charge)

We note that for an infinite sheet of charge $\mathbf{E}$ is the same at all points above the x–y plane, and a similar statement applies for points below the x–y plane.

Concept Question 4-4: When characterizing the electrical permittivity of a material, what do the terms *linear* and *isotropic* mean?

Concept Question 4-5: If the electric field is zero at a given point in space, does this imply the absence of electric charges?

Concept Question 4-6: State the principle of linear superposition as it applies to the electric field due to a distribution of electric charge.

Exercise 4-6: An infinite sheet with uniform surface charge density ρ_s is located at $z = 0$ (x–y plane), and another infinite sheet with density $-\rho_s$ is located at $z = 2$ m with both in free space. Determine $\mathbf{E}$ everywhere.

Answer: $\mathbf{E} = 0$ for $z < 0$; $\mathbf{E} = \hat{\mathbf{z}}\rho_s/\varepsilon_0$ for $0 < z < 2$ m; and $\mathbf{E} = 0$ for $z > 2$ m. (See EM.)

4-4 Gauss's Law

In this section, we use Maxwell's equations to confirm the expressions for the electric field implied by Coulomb's law, and propose alternative techniques for evaluating electric fields induced by electric charge. To that end, we restate Eq. (4.1a):

$$\nabla\cdot\mathbf{D} = \rho_v, \tag{4.26}$$

(differential form of Gauss's law)

which is referred to as the differential form of ***Gauss's law***. The adjective "differential" refers to the fact that the divergence operation involves spatial derivatives. As we see shortly, Eq. (4.26) can be converted to an integral form. When solving electromagnetic problems, we often go back and forth between equations in differential and integral form, depending on which of the two happens to be the more applicable or convenient to use. To convert Eq. (4.26) into integral form, we multiply both sides by $d\mathcal{U}$ and evaluate their integrals over an arbitrary volume $\mathcal{U}$:

$$\int_{\mathcal{U}} \nabla\cdot\mathbf{D}\,d\mathcal{U} = \int_{\mathcal{U}} \rho_v\,d\mathcal{U} = Q. \tag{4.27}$$

Here, Q is the total charge enclosed in $\mathcal{U}$. The divergence theorem, given by Eq. (3.98), states that the volume integral of the divergence of any vector over a volume $\mathcal{U}$ equals the total outward flux of that vector through the surface S enclosing $\mathcal{U}$. Thus, for the vector $\mathbf{D}$,

$$\int_{\mathcal{U}} \nabla\cdot\mathbf{D}\,d\mathcal{U} = \oint_S \mathbf{D}\cdot d\mathbf{s}. \tag{4.28}$$

Comparison of Eq. (4.27) with Eq. (4.28) leads to

$$\oint_S \mathbf{D}\cdot d\mathbf{s} = Q. \tag{4.29}$$

(integral form of Gauss's law)

▶ The integral form of Gauss's law is illustrated diagrammatically in **Fig. 4-8**; for each differential surface element $d\mathbf{s}$, $\mathbf{D}\cdot d\mathbf{s}$ is the electric field flux flowing outward of $\mathcal{U}$ through $d\mathbf{s}$, and the total flux through surface S equals the enclosed charge Q. The surface S is called a ***Gaussian surface***. ◀

The integral form of Gauss's law can be applied to determine $\mathbf{D}$ due to a single isolated point charge q by enclosing the latter with a closed, spherical, Gaussian surface S of arbitrary radius R centered at q (**Fig. 4-9**). From symmetry considerations and assuming that q is positive, the direction of $\mathbf{D}$ must be radially outward along the unit vector $\hat{\mathbf{R}}$, and D_R, which is the magnitude of $\mathbf{D}$, must be the same at all points on S. Thus, at any point on S,

$$\mathbf{D} = \hat{\mathbf{R}}D_R, \tag{4.30}$$

等式。 (4.23) 并且将 b 替换为 r，我们得到以下由环引起的场的表达式：

$$d\mathbf{E} = \hat{\mathbf{z}}\,\frac{h}{4\pi\varepsilon_0(r^2+h^2)^{3/2}}\,(2\pi\rho_s r\,dr).$$

P 处的总场通过在 r = 0 到 r = a 的范围内积分表达式来获得：

$$\mathbf{E} = \hat{\mathbf{z}}\,\frac{\rho_s h}{2\varepsilon_0}\int_0^a \frac{r\,dr}{(r^2+h^2)^{3/2}} = \pm\hat{\mathbf{z}}\,\frac{\rho_s}{2\varepsilon_0}\left[1-\frac{|h|}{\sqrt{a^2+h^2}}\right], \tag{4.24}$$

当 h > 0（P 在盘上方）时为加号，当 h < 0（P 在盘下方）时为负号。

对于具有 = ∞ 的无限电荷片，

$$\mathbf{E} = \pm\hat{\mathbf{z}}\,\frac{\rho_s}{2\varepsilon_0}. \tag{4.25}$$

(infinite sheet of charge)

我们注意到，对于无限电荷片，E 在 x-y 平面上方的所有点上都是相同的，并且类似的陈述适用于 x-y 平面下方的点。

概念问题 4-4：在表征材料的介电常数时，术语“线性”和“各向同性”意味着什么？

概念问题 4-5：如果空间中给定点的电场为零，这是否意味着不存在电荷？

概念问题 4-6：陈述线性叠加原理，因为它适用于电荷分布产生的电场。

练习 4-6：具有均匀表面电荷密度 ρ_s 的无限片位于 z = 0（x-y 平面），另一个具有密度 $-\rho_s$ 的无限片位于 z = 2 m，两者都在自由空间中。处处求 E。

答案：E = 0 代表 z < 0； $\mathbf{E} = \hat{\mathbf{z}}\rho_s/\varepsilon_0$ 为 0 < z < 2 m；和 E = 0 对于 z > 2 m。(参见EM。)

4-4 高斯定律

在本节中，我们使用麦克斯韦方程组来确认库仑定律所隐含的电场表达式，并提出评估电荷感应电场的替代技术。为此，我们重申方程。 (4.1a)：

$$\nabla\cdot\mathbf{D} = \rho_v, \tag{4.26}$$

(高斯定律的微分形式)

这被称为高斯定律的微分形式。形容词“微分”指的是散度运算涉及空间导数。正如我们很快看到的，等式。 (4.26) 可以转化为积分形式。在解决电磁问题时，我们经常在微分形式和积分形式的方程之间来回切换，具体取决于两者中哪一个更适用或更方便使用。转换方程。将式（4.26）转化为积分形式，我们将两边乘以 dυ 并计算它们在任意体积 υ 上的积分：

$$\int_{\mathcal{U}} \nabla\cdot\mathbf{D}\,d\mathcal{U} = \int_{\mathcal{U}} \rho_v\,d\mathcal{U} = Q. \tag{4.27}$$

这里，Q 是包含在 υ 中的总电荷。散度定理，由方程给出。 (3.98) 指出，任何向量在体积 υ 上的散度的体积积分等于该向量通过包围 υ 的表面 S 的总向外通量。因此，对于向量 D，

$$\int_{\mathcal{U}} \nabla\cdot\mathbf{D}\,d\mathcal{U} = \oint_S \mathbf{D}\cdot d\mathbf{s}. \tag{4.28}$$

等式的比较（4.27）与方程。 (4.28) 导致

$$\oint_S \mathbf{D}\cdot d\mathbf{s} = Q. \tag{4.29}$$

(高斯定律的积分形式)

高斯定律的积分形式见图4-8。对于有向面元 ds，D·ds 是沿曲面外法向穿过面元的电通量。闭合曲面 S 上的总电通量等于其中包围的电荷 Q；S 称为高斯面。

高斯定律的积分形式可用于确定由单个孤立点电荷 q 引起的 D，方法是用一个以 q 为中心、任意半径 R 的闭合球形高斯表面 S 包围后者（图 4-9）。从对称性考虑，假设 q 为正，D 的方向必须沿着单位向量 R̂ 径向向外，并且 D_R（即 D 的大小）在 S 上的所有点上必须相同。因此，在 S 上的任何点，

$$\mathbf{D} = \hat{\mathbf{R}}D_R, \tag{4.30}$$